

Extra study and consolidation in Algebra

Thobias Høivik

Contents

1 What is this?	3
2 Repeating some stuff about ideals	3

1 What is this?

Due to my being very interested in algebra, and interested in becoming very good at it, I have these extra notes for me to work through a lot of problems. Basically, there will probably not be any pedagogical value here.

2 Repeating some stuff about ideals

Problem 2.1. Let

$$R = \frac{k[x, y]}{(xy)}$$

and write $\bar{x}, \bar{y}$ for the residue classes.
What is $(\bar{x})$ as an ideal of R ?

Compute

$$R/(\bar{x}), \quad R/(\bar{y}), \quad R/(\bar{x}, \bar{y}).$$

Then classify them as prime/maximal/neither.

Sol. We note that the ideal generated by $\bar{x} = x + (xy)$ is

$$(\bar{x}) = \{\overline{xf} \mid f \in k[x, y]\}.$$

We have $(xy) \subseteq (x)$ and $(xy) \subseteq (y)$ as ideals of $k[x, y]$.

Consider the projection $\pi : k[x, y] \rightarrow R$ by

$$\pi(a) = \bar{a} = a + (xy).$$

Since π is surjective and $\{x\} \subseteq k[x, y]$ we have

$$\pi[(x)] = (\pi x) = (\bar{x}),$$

(note: surjectivity guarantees that scalar multiplication in R can be pulled back to $k[x, y]$, ensuring $\pi[(x)] \supseteq (\pi(x))$) but also

$$\pi[(x)] = \frac{(x)}{(xy)}$$

since $(xy) \subseteq (x)$.

Similarly,

$$\frac{(y)}{(xy)} = \pi[(y)] = (\bar{y}),$$

and

$$\frac{(x, y)}{(xy)} = \pi[(x, y)] = (\bar{x}, \bar{y}).$$

Thus we can apply the third isomorphism theorem to get

$$\frac{R}{(\bar{x})} = \frac{k[x, y]/(xy)}{(x)/(xy)} \cong \frac{k[x, y]}{(x)} \cong k[y],$$

and

$$\frac{R}{(\bar{y})} = \frac{k[x, y]/(xy)}{(y)/(xy)} \cong \frac{k[x, y]}{(y)} \cong k[x],$$

and

$$\frac{R}{(\bar{x}, \bar{y})} = \frac{k[x, y]/(xy)}{(x, y)/(xy)} \cong \frac{k[x, y]}{(x, y)} \cong k.$$

The first two are integral domains so $(\bar{x}), (\bar{y})$ are prime, though not maximal, while $(\bar{x}, \bar{y})$ is maximal since $R/(\bar{x}, \bar{y}) \cong k$ is a field. $\square$

Problem 2.2. Let

$$R = \frac{k[x, y]}{(xy)},$$

and let

$$\pi : k[x, y] \rightarrow R$$

be the quotient map.

Prove that every prime ideal $P \subseteq R$ contains $\bar{x}$ or $\bar{y}$.

Then, classify every prime ideal of R .

You may use without proof:

$$P \subseteq R \text{ prime} \Leftrightarrow \pi^{-1}(P) \subseteq k[x, y] \text{ is prime containing } (xy).$$

Proof. We have $\overline{xy} = 0$, and since P is an ideal, and thus a group, we have $0 \in P$, and since P is prime we must then have

$$\bar{x} \in P \text{ or } \bar{y} \in P.$$

Suppose P contains $\bar{x}$. Then, $(\bar{x}) \subseteq P$.

Then, by the third isomorphism theorem,

$$\frac{R/(\bar{x})}{P/(\bar{x})} \cong R/P,$$

but

$$\frac{R}{(\bar{x})} \cong k[y]$$

so

$$R/P \cong \frac{k[y]}{P/(\bar{x})}.$$

Since P is a prime ideal of R , the quotient R/P is an integral domain. Therefore, $P/(\bar{x})$ must be a prime ideal of $k[y]$.

Since $k[y]$ is a Principal Ideal Domain (PID), its prime ideals are:

1. The zero ideal (0) , or
2. Irreducible polynomial ideals $(f(y))$, where $f(y) \in k[y]$ is an irreducible polynomial.

Under the canonical quotient map $R \rightarrow R/(\bar{x}) \cong k[y]$, these correspond precisely to the prime ideals of R containing $\bar{x}$:

1. $P = (\bar{x})$, corresponding to $(0) \subset k[y]$.
2. $P = (\bar{x}, f(\bar{y}))$, where $f(y) \in k[y]$ is irreducible.

By symmetric reasoning, if $\bar{y} \in P$, then $P/(\bar{y})$ is a prime ideal of $k[x]$, which yields:

1. $P = (\bar{y})$, corresponding to $(0) \subset k[x]$.
2. $P = (\bar{y}, g(\bar{x}))$, where $g(x) \in k[x]$ is irreducible.

Combining both cases, the prime ideals of R consist of:

- Minimal prime ideals: $(\bar{x})$ and $(\bar{y})$.
- Maximal ideals: $(\bar{x}, f(\bar{y}))$ for irreducible $f(y) \in k[y]$, and $(\bar{y}, g(\bar{x}))$ for irreducible $g(x) \in k[x]$.

□